

Workshop: Data Analytics



U.S. Hotel Dataset

Data Cleaning Workshop

From Raw Data to Business Insights

Part 1: Data Cleaning & Preparation

Identify. Standardize. Merge.



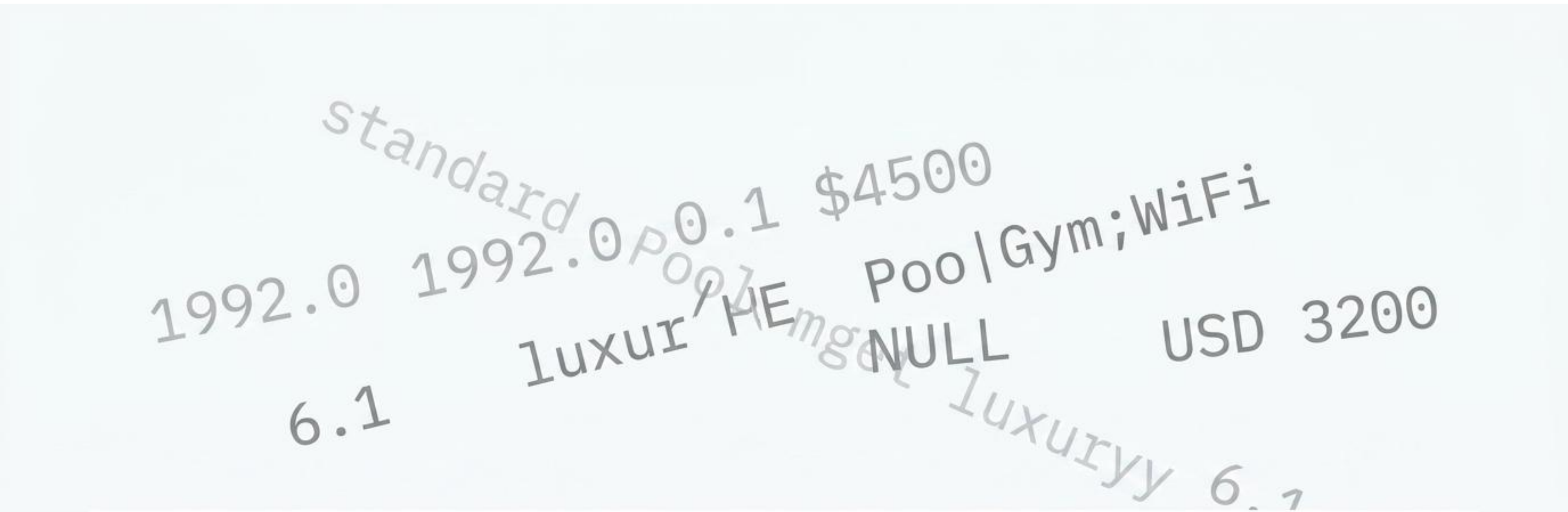
hotel.csv

Part 2: Exploration & Visualization

Analyze. Discover. Decide.



AI Agent



Real-World Data is Rarely Perfect

Before building analytics or AI applications, critical quality issues must be resolved: missing values, duplicate records, format errors, and fragmented files.

Dataset A

hotels_info.csv

Focus: **Identity & Specs**

Hotel ID
Hotel Name
City
State
Address
Category
Star
Year Opened
Number of Rooms

Dataset B

hotels_pricing.csv

Focus: **Metrics & Value**

Hotel ID
Price per Night
Rating
Availability
Number of Reviews
Discount Percentage
Amenities
Peak Season

The Common Thread: **hotel_id**

Understanding the Data

`df.info()`

Assess structural integrity, row counts, and data types.

`df.head()`

Spot-check sample records for obvious human errors.

`df.isnull().sum()`

Identify critical missing variables and data gaps.

`df.duplicated()`

Detect repeated records to ensure statistical accuracy.

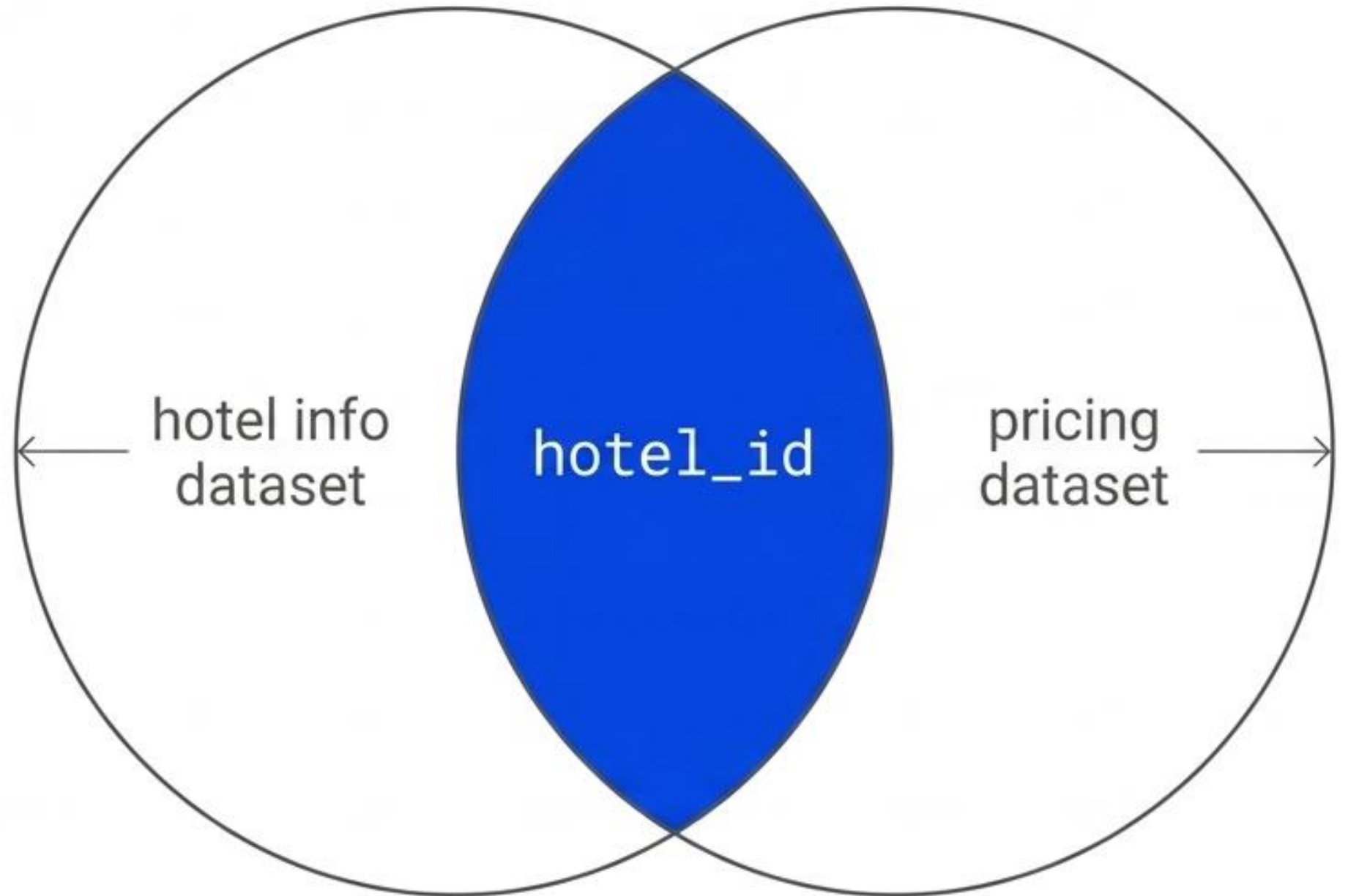
The Refinement Process

Symptom	Diagnosis	Treatment	Cure
USD 3200 & 150 rooms	Text hidden in numeric fields	Strip text/symbols	3200 & 150
MIAMI & stanndard	Inconsistent formatting	Title case / Mapping dictionary	Miami & Standard
Rating 6.1	Business rule violation (Max 5)	Replace with Null	NULL
Pool Gym;WiFi	Mixed separators	Standardize delimiter	Pool, Gym, Wi-Fi
3 stars	Text hidden in numeric fields	Parse digit	3

Forging a Complete Record

We perform an Inner Join to merge the datasets, keeping only hotels that exist in both files.

- Drop nameless records (hotel_name = NULL)
- Remove statistical duplicates



AI systems and analytics require complete, verified records to function accurately.

The Clean Foundation



42		The Phoenix Palace		Phoenix		Budget		268		4349.37
11		Grand Los Angeles Hotel		Los Angeles		Luxury		270		4888.67
22		The Chicago Palace		Chicago		Luxury		201		4170.37

The final output (hotels.csv) is a singular, normalized dataset ready to power analytics and the AI Hotel Search Tool.

Workshop: Data Cleansing & Data Preparation

Set Up

1. Go to Google Drive and create a new folder named **“Day28_Workshop”**
2. Download 3 files in Google Classroom
 - [Tutorial] Hotel_Data_Workshop_Colab.ipynb
 - hotels_info.csv
 - hotels_pricing.csv

Then upload all files into your drive (../MyDrive/Day28_Workshop)
3. Open your copied notebook in Google Colab and follow the Data Cleansing tutorial — run each cell in order

```
MyDrive/  
└─ Day28_Workshop/  
    └─ [Tutorial] Hotel_Data_Workshop_Colab.ipynb  
    └─ hotels_info.csv  
    └─ hotels_pricing.csv
```